Overview of Wi-Fi-based Automatic Passenger Counting Solutions in Public Urban Transport

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Abstract. Automatic Passenger Counting (APC) solutions imply different ways of counting passengers and the most common division of APC is on independent and integrated systems. This paper analyzes independent passenger counting solutions based on Wi-Fi technology. In this paper the researched problem is reflected in the review of different approaches to ridership via Wi-Fi technology in Public urban Passenger Transport (PPT), and the most common possible division is on active and passive scanning of mobile user devices for probe request frames. Passenger counting using Wi-Fi-based APC in urban traffic can be broken down (classified) according to several observation factors, such as according to the location of the counting (inside and outside the vehicle), according to the observed means of transport (bus, train/metro, etc.), according to the purpose of the collected data, estimates, actual numerical values and methods/methodology of passenger counting. This paper can serve as a contribution to defining the taxonomy of the way passengers are counted using Wi-Fi technology in different subsystems of public urban passenger transport as a part of future development or ongoing integration into a Smart City concept.

Keywords: Automatic Passenger Counting (APC), Wi-Fi, Active scanning, Passive scanning, Probe request frame, Public Urban Passenger Transport (PPT), Wi-Fi based APC, Ridership, Smart City.

1 Introduction

In the twentieth and twenty-first century, the major societal change that characterized that period was the trend of urbanization (the displacement of population from rural to urban environments). This process created a new demand for transportation and generated new ideas such as urban mobility, people mobility analytics, Smart Cities and many others. Urban mobility is increasing proportionally to the development of the urban environment, but in some parts of the world it is even faster and developing exponentially. This is caused by an increased number of people and citizens in certain urban environments that sometimes generate faster than the city infrastructure can follow. This increasing number of people in the cities creates an ever-increasing demand for faster, cheaper, more comfortable, safer and more ecologically conscious

urban transport. And so, public transport is the primary mode of transport for a large number of people worldwide. However, there is still a large proportion of the global population who use passenger cars as their primary mode of transportation. For example, most people in the European Union use their personal cars as their primary mode of transportation (4 trillion passenger kilometers per year) and bus and tram/metro are less than an eighth of that (569 billion passenger kilometers for the bus and 89 billion for tram/metro) [1]. This means that car traffic accounted for 71% of the total share of passenger mileage in the EU for 2018, while the bus occupied 8% and the tram/metro accounted for 1.4% [2].

In response to that, increasing development and further optimization processes for PPT are designed and created (alongside urban housing, business and commercial hubs and districts) as an important component in the expansion of cities and urban environments. It can be argued that transportation is the most important link in the urban environment chain and its proper operation is of the utmost importance for the city to function. In the future, the ratio of using personal cars and PPT vehicles will have to change significantly if there is going to be an economical, ecological and sustainable future, making the Smart City concept a reality and for the transition into the Society 5.0. Even so, the study [3] from 2014 shows that the largest percentage of passengers in the European Union, 59% of them, use some form of public passenger transport (PPT) - tram, bus, metro etc., in their everyday life [3].

To decrease the usage of personal transportation in favor of the public transportation, one needs to understand the problems of urban city traffic. For this to happen, the officials in charge need to understand that dealing with the increasing traffic and traffic congestion does not necessarily mean increasing the infrastructure capacities, but can mean transferring a part of the people from personal transportation to public transportation, if only to commute (peak rush hours). The best way to accomplish this is to optimize the public urban transport network so it is a viable option over personal transportation. Creating a viable alternative means having a great understanding of the PPT network and people's mobility to design and improve the system in real time. One of the best ways to increase the number of passengers who use PPT is to optimize the experience and travel times and the best way to do so is to use Automatic Passenger Counting (APC) to gather data in real time.

Any form of counting transport and transported entities should be the backbone of traffic and transport and the information gathered is used to design, create, reconstruct and optimize the transport network or infrastructure. The counting data can be used to notice certain patterns in behavior for certain locations and time periods which can then be used to redistribute resources and personnel to where they are needed in accordance with the overall goals and purposes. The correct assessment or a precise number of passengers in PPT is of significance for proper management and organization of public urban transport. There are many different methods of counting passengers in public transport, as shown in paper [4], but this paper has focused on the Wi-Fi-based ones that can vary in method, mounting locations (inside the vehicle or in the

station) and can provide various data to be used for different purposes (ridership, OD matrix, driving schedule, any form of people mobility analytics - PmA, etc.). Wi-Fi is not the only wireless technology (Bluetooth, ZigBee, etc.) that can be used for this purpose, but it has been chosen because many transport service providers already have some form of Wi-Fi capability in their PPT vehicles. Also, a great advantage of APC solutions based on Wi-Fi is the increasing number of smartphone users worldwide (from 1 billion in 2012 to 3.8 billion in 2021) [5].

The goal of the research was to provide an overview of automatic passenger counting solutions based on Wi-Fi in public city transport, i.e. to determine different approaches to passenger counting using Wi-Fi technology. The motive for researching these topics is related to the unavailability of the APC taxonomy, i.e. the clear division and applicability of the method of counting passengers using Wi-Fi technology in various subsystems of public urban transport, as well as the widespread usage of integrated Wi-Fi networks in PPT vehicles worldwide. Whilst investigating the problem of counting passengers via Wi-Fi, the problem of inconsistency of terminology related to active and passive counting methods was noticed, which is described further in the paper.

The second section provides an overview and review of the existing literature and research in the analyzed area. The third section describes the features of Wi-Fi as the basis for APC solutions. Furthermore, different scanning methods that represent the terminology problem are described and reviewed in the context of passenger counting. In the fourth section analysis of the results of existing research is presented according to several different factors, such as the location of the counting (in the vehicle/outside the vehicle), the observed means of transport (bus, train, metro, etc.), the purpose of the collected data, estimation, real counted values and methods/methodology of passenger counting. It is explained why some information was separated from reviews and what are the major connections that link most of the conducted experiments. In section five, all the findings and conclusions of the research were synthesized and guidelines and plans for future research were given.

2 Existing research

The authors of [6] researched and experimented with the concept of passenger counting based on Wi-Fi sensing system (media access control or MAC address collecting) and signal strength analysis with camera-based APC for reviewing and checking results. Their research was conducted at four bus stops along the main transit corridor in Charlottesville, USA. The results they gathered proved that their solution has a precision of 91% for bus passengers and that the waiting time had an error rate of only 7 seconds. They concluded that their system can be a viable and low-cost solution with high potential with Wi-Fi sensing. Paper [7] has proposed statistical methods for estimating OD matrices for public transport trip level based on collected Wi-Fi data. The authors have installed their equipment in 32 buses in the city of Stavanger in

Norway and have collected Wi-Fi data (probe request frames and time between the first and the last frame) for a period of several months and based on those data they have analyzed the probability for each detected device to be on or off the bus and estimated the OD matrix. There was no base truth data for the OD matrix so some small errors in estimates are possible. Using data from collecting probe request frames is the basis of research [8] and the case study was conducted in the city of Obuse in Japan inside a "hop-on-hop-off" type of bus. The goal was to estimate bus passenger volume using a Wi-Fi scanner that collected MAC addresses and global positioning system (GPS) data and clean the raw data enough to perform the estimation effectively. The authors have concluded that the correlation between real values and their estimation is high and amounts to 0.78. They continue to describe the process for collecting and processing said data to generate the OD matrix and speed calculations to determine if the MAC address timestamp data and GPS logs correlate to the actual bus speed. The problems they voice are the number of Wi-Fi capable devices that are almost always a little wrong because some passengers carry more than one device and some might not have any and some devices can change their MAC address after a software upgrade. Another limitation they encountered was the large range Wi-Fi sensors have and so, a lot of noise is generated in the collected datasets. This can of course be countered by using a directional antenna or by lowering the effective covering range.

Authors of [9] state that their solution is low-cost infrastructure able to collect geolocation data and provide services with the goal of optimizing the routes/lines and driving schedules by providing real-time information. Their devised system was deployed in nineteen PPT vehicles (buses), on the Madeira Island in Portugal, to collect Wi-Fi data which they then analyzed and compared to, what they considered, ground truth data gathered from ticket validation. They also used the collected data to estimate the numbers of passengers that exit/enter, detect peak values and OD matrices. The idea, again, is to use the existing Wi-Fi infrastructure to collect MAC addresses and geolocation data, processed to exclude random MAC addresses by filtering those that come up just once per trip, for mapping bus routes and stops and to log bus trips. The drawback of the proposed solution is the passive nature of collecting probe frames that is dependent on the AP's frequency. However, this solution can be adapted for monitoring crowd levels, disruptions, alerts, waiting times and lines, etc. The value of Wi-Fi data and smartphone devices in everyday life is also shown in paper [10]. The authors attempt to measure the correlation between counting Wi-Fi probe request frames and the real number of people. Also, they compare their results with manual counting results conducted for eight hours and conclude that the results are promising and the correlation high (0.89322). They also conclude that the used method is adequate for estimating the number of people, even though it cannot reliably provide the exact number and that it can be used to calculate and estimate the waiting times in correlation with passengers joining or leaving the crowd. However, the authors have not conducted their research in public transport vehicles or stations, but rather in a large hall with a large crowd of people.

Paper [11] describes a solution with Wi-Fi sensor logs from smartphone devices and proposes an unsupervised “Passively Encountering Group” (PEG) detection framework. Every smartphone in the vicinity of Wi-Fi AP is detected and the signal strength is measured and logged. The log is then sent to the server where the information is used to form PEGs. The developed model identifies the subjects in the proximity of the AP based on the strength of the Wi-Fi signal and then has a score attached to it based on their metric system to determine the level of proximity. The experiment was conducted on the university campus of the Indian Institute of Technology in the city of Kharagpur and included over twenty-five subjects for a time period of six months. The conclusion was that the precision for this solution is higher than 90% and the authors state that they have identified two problems in their experiment, which are identification of subjects in the proximity and detecting groups from the proximity information. Another research experimenting with signal strength in Wi-Fi networks [12], has a proposed solution for estimating the number of people based on a regression-based system. The research was conducted in a controlled environment in the Kindai University laboratory in Osaka using a Wi-Fi AP and a node (Raspberry Pi 2 Model B with a USB WiFi module). The authors state that the precision is higher using a support vector regression-based system rather than the linear regression-based one. They also conclude that the precision of the support vector regression-based system for estimating people is 0.772, for the degree of congestion it is 0.946 and for the presence or absence of people it is 0.982, but they state that they need to evaluate the proposed solution in different environments such as large halls, outdoors and other similar places, as well as to increase the number of used APs.

The authors of [13] describe and analyze their proposed solution for monitoring and predicting the movements of people using, what they call, a passive Wi-Fi sensing system. However, their experiment was also performed in enclosed locations, in two different offices in Singapore, and the authors state that this is because of the open space and dynamic environments. The main proposition is the estimation of the Length Of Stay (LOS) in real time and is based on collecting probe requests and other types of frames. They use a linear Support Vector Machine (SVM) classifier and an online Stochastic Gradient Descent (SGD) mechanism to account for the dynamic environment. The conclusion of this experiment is that SGD is optimal for dynamic and rapidly changing environments and SVM is more precise in less dynamic ones. The results from this are: SVM - no more than three seconds and SGD - less than one second to train on average. The solution proposed by the authors [14] is based on estimating the passenger occupancy and flow in buses using a mobile application capable of detecting other mobile devices over Bluetooth and Wi-Fi. They state that, using a binary classification model, they can determine whether a bus is full or not based on the number of Wi-Fi users. The application continuously scans for devices and counts how many it has detected and then sends the data to the server to be processed. The experiment was performed in the city of Dublin in Ireland, with volunteers who agreed to install and use their application for one week. They would then activate the application once on the bus and would count the real value to submit to the system. Afterwards, the authors developed their model and concluded that it has

an almost 88% precision in the performed tests. Also, the authors of [15] proposed a feasibility study to estimate the bus passengers and congestion they generate by detecting Wi-Fi probe requests in buses and at bus stops. The experiment was conducted at the bus stop in the Ritsumeikan University in Japan and on board the bus at the local train station. The data were collected at the bus stops as well as on board buses during peak hours. The analysis performed using the Pearson correlation showed the correlation of 0.6215 and constant of proportionality, devices/passengers, around 5.5. They state that congestion might also be estimated but the collected data were insufficient. This method might also be used to show trends in behavior of passengers and their daily routine. One more paper that proposes solutions for estimating the number of passengers/Wi-Fi devices via probe request collecting is [16]. The stated purpose of the experiment was to avoid the problem of random MAC addresses by using the propagated information with no need for the real MAC address. The paper was a continuation of an already conducted experiment [17] that had analyzed the probe request process in detail. The proposed solution was modeled to detect Wi-Fi devices when they enter/exit and while they are in the detection zone and the experiment was conducted in the laboratory conditions at the Federal University of Rio de Janeiro. The analyzed solution has shown high correlations (Herlock 0.087, LR-based method 0.383, SVR-based method 0.298) between the number of mobile devices and the real number of people/passengers. Also, Pearson's correlation calculation has shown high correlation as well - 0.896, with a low mean relative error - 0.087.

3 Features of Wi-Fi as the basis of an APC solutions

The most reliable information on the actual number of passengers in vehicles is obtained based on the number of arrivals and departures of passengers at each station along the public transport line. Based on this, highly reliable information can be collected and then used to determine the distribution of trip times and efficiency of individual lines and vehicles, optimize the timetables, analyze the vehicle routes and lines, increase or shorten routes, add or remove certain stations, etc. However, conducting such experiments requires significant financial investment, as well as other resources (human and time), so it is necessary to harmonize data collection in accordance with the capabilities of the subjects requesting the measuring. Traffic counting, i.e. passenger counting is carried out by manual or automatic methods and the choice primarily depends on the type of information collected, the duration of the count, and the available financial budget. Although manual counting methods are still used, efforts are being made to automate passenger counting to reduce the need for human components in the process. The reason is the unreliable and unpredictable nature of the people in the counting process, given that it is subject to subjective "circumstances" (fatigue, distraction, forgetfulness, etc.).

It is the automatic counting of passengers that eliminates "subjective" circumstances, which in turn eliminates the problem of the human error factor that affects the collection of data. There are currently a number of commercial and "improvised" automatic counting solutions available for the implementation as seen in the existing research

that are further explained in Section four. Nevertheless, globally, their application in PPT vehicles is still not widely applied and the reasons are multiple, but boil down to the fact that a large number of PPT service providers do not have the financial power to procure automated counters and there is no universal solution that is applicable in all situations. The absence of universal solutions means different public service providers use different means of transport that operate in different, variable conditions (different vehicle dimensions and types, unpredictability of traffic in urban areas, etc.). Still, pilot experiments with commercial and "improvised" solutions are being conducted with increasing interest to find an efficient and reliable way to collect passenger data. Furthermore, the main feature of the APC solutions is manifested in the fact that they are able to respond to the imposed demands caused precisely by the unpredictability and dynamic nature of urban traffic as well as any other environment in which they operate. Also, when passenger counting is performed, it is unnecessary to implement the APC components in all the transport vehicles in the fleet. The counting is performed in "rotation", meaning that the vehicles with counters can be moved and swapped on other lines if needed. This implies that counting components are removed and implemented in individual JGP lines as needed, and it is estimated that approximately 10% of the complete vehicle fleet has APC components implemented in vehicles at any given time as described in report [18].

Paper [4] presents an overview of the most used APC solutions and it was noticed that, largely represented, counting methods are based on Wi-Fi technology. It was concluded that the reason for that is the widespread use of devices that support 802.11 standards (2018. 22 billion and predictions are that in 2025 the number will grow to 38.6 billion and in 2030 to 50 billion) and they are used to represent and determine the number of passengers in JGP vehicles [19]. Therefore, the identified area of interest is based on Wi-Fi technology and is shown in Fig. 1.

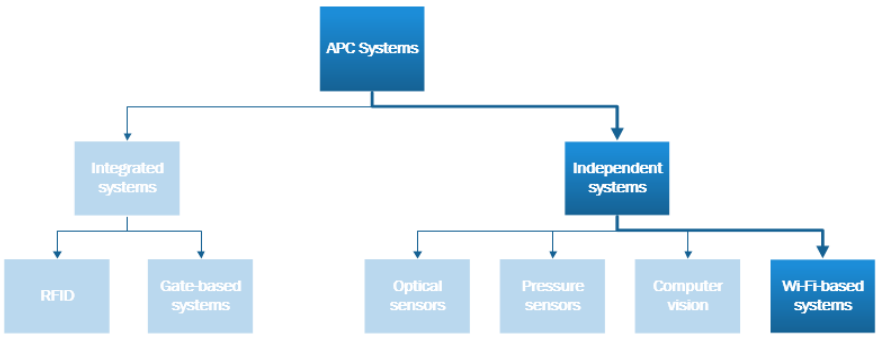

Fig. 1. Area and scope of research

As mentioned, APC solutions based on Wi-Fi technology do not have universal application in PPT vehicles and one of the reasons that was determined in pilot experiments was the vast pool of data that can be gathered and all the different possibilities offered by such a solution. So, certain characteristics and collected data from 802.11

devices that are exploited to determine the number of passengers, i.e. devices in the vehicle, will be defined further in the section and the concept of operation for APC solutions based on Wi-Fi technology will be described.

3.1 WLAN overview

A fundamental requirement for modern digital society, especially for concepts like Smart City and Society 5.0, is the reliability and availability of information any time and any place. Such requirements raise the need for certain standards of mobility, flexibility, ease of implementation, reasonable prices, security and unobtrusiveness, which are a large part of the WLANs [20]. In accordance with that, the main goal of WLAN is to provide mobile and secure access to a network for mobile devices or users via wireless network media. With that said, counting mobile devices or users via Wi-Fi technology is based on the process of collecting probe request frames in the association of service communication between devices (Machine-to-Machine communication - M2M). This is why it is necessary to first explain this process as an essential part of the Wi-Fi-based counting process. The association service is a process in 802.11 M2M communication for registering a user device with an access point (AP) in order to establish a connection and gain access to the internet via the wireless network. To be able to send and receive data over the AP, the user device must be authorized and authenticated via the process of association. In order to establish a viable M2M communication, a number of frames must be exchanged between the user device and the AP, the first of which is a probe request frame. This is the most important frame with regard to collecting data for counting because it is transmitted frequently and with no input from the user, but has some useful data stored in it (MAC address, timestamp, SSID, IBSS, etc.) [21].

The process of association can begin when the network is identified by the user device while scanning the surrounding environment. This mostly happens automatically, as was mentioned, because it transpires frequently and with no user input or interference. However, this process can be divided into active and passive modes, depending on the user device sending the probe request frames or waiting to receive beacon frames. So, active scanning is the process in which a user device transmits a probe request frame on every channel and then waits for the response from the nearby AP. The 802.11 standard has no fixed time defined for how long the user device has to wait for the response from the AP before sending a new batch of probe request frames. However, the usual time period is estimated to be 10 milliseconds. A simplification of the aforementioned process is shown in Fig. 2 [22].

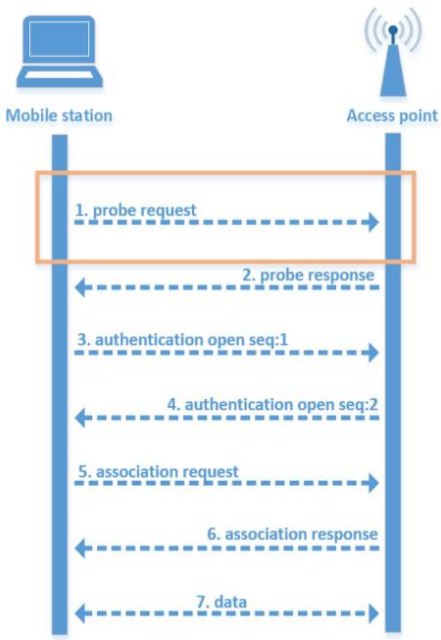

Fig. 2. Process of generating probe request frames and the focus for passenger counting

Passive scanning, on the other hand, is a process in which the user device does not search for the nearby AP or initiate the communication, but listens instead for the transmitted beacon frames on available channels. These beacon frames are being transmitted by the AP every 100 milliseconds which makes it possible for the user device to miss them because it had not listened for long enough on the right channel. This is why active scanning can be considered better for the process of collecting data [22].

3.2 Concept for collecting data from PPT using Wi-Fi

With everything described in the last section, the active scanning method is better for counting passengers in PPT and, as mentioned earlier, the probe request frames that are transmitted every 10 milliseconds are filled with potentially useful data. However, in the research for this paper the authors have observed that some solutions use probe request frame collecting for their raw data gathering, but keep referring to it as passive scanning. The reason for this terminology difference is because some researchers and authors [8], [9], [10], [13] consider their scanning to be unobtrusive for the passengers and so consider it passive. This conflict in terminology is probably caused by ambiguity since it is active from WLAN and network security perspective, but unobtrusive, non-invasive and indirect, i.e. passive from the passenger counting perspective. For the purpose of this paper and counting passengers in PPT, the probe request frame collecting will be considered a passive method because of the unobtrusive nature towards the users.

Taking this into consideration, passenger counting can be performed at several levels depending on the purpose for which the data are collected. Therefore, it can be divided into:

- System level - making decisions on budgets and finance planning;
- Route or line level - planning and scheduling the means of transport;
- Travel level - determining whether to add or remove a specific PPT line;
- Stop level - determining if and where to create a station;
- Segment level - planning the transportation service [18].

In cases of smaller service areas or when it comes to a smaller PPT service provider, passenger counting is done at the "*macro*" level. This implies a systematic and route/line approach. However, the greater the complexity of the PPT infrastructure, the more passenger providers count at the "micro" levels (more precisely, previously mentioned travel, stop and segment levels). In addition, the more detailed the "micro" levels are, the more demanding the number of passengers and the greater the need for technical and human resources support. Furthermore, based on examined existing research [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] some pros and cons were identified. Pros of Wi-Fi based APC solution are:

- Affordable prices of components;
- Ever-growing number of Wi-Fi capable devices in everyday use;
- Forgetfulness of users to deactivate their Wi-Fi capability;
- Unobtrusiveness;
- Does not make passengers feel monitored;
- Large amount of information provided by probe request frame.

Despite stated advantages, some disadvantages were also identified:

- Difference in the number of passengers and user devices;
    - Passenger can have just one device;
    - Passenger can have more than one device;
    - Passenger can have no devices;
    - Detection of devices outside the vehicle or designated location.
- Need for calibration because of the difference in passengers and devices;
- Randomization of MAC addresses;
- Variations in sending probe request frames;
- Dynamic nature of PPT environment;
    - Variable distances of PPT stops and stations;
    - Variable waiting time of PPT vehicles;

Also, there are many different uses for data collected in the previously described ways in accordance with the mentioned research, ridership being the primary one, but with all the valuable data gathered, other uses can be extrapolated - OD matrices, waiting times, peak hours and behavioral trend, degree of congestion, etc. Most of that data can be formulated in some form of PmA results that help discover and analyze trends in passenger behavior. This, of course, greatly helps in designing, creating, running, optimizing and maintaining PPT systems, infrastructure, networks and vehicles.

3.3 Wi-Fi-based APC

As already established, the most practical way to collect data with an APC based on Wi-Fi is to use active scanning because of the previously described probe request frame collection process that allows for flexibility when defining whether or not the user devices are inside the vehicle. For example, automatic probe request frame counting solution is practical in cases where the flow of people is dynamic, i.e. where people are passing by or staying just temporarily (public transport vehicles) because it has a relatively low probe request transmission time and can be manually adjusted if needed. It is also good for use in cases where the flow of people is static, i.e. where people stay in one place for a longer period of time (long-distance travel, PPT stations, etc.). In these circumstances it is predicted that their devices will actively seek AP access.

Passive scanning, on the other hand, is quite impractical for counting people in general, because of its slow beacon frame transmission rate and channel distribution. This means that the counting error and reliability of the system suffer greatly and can be increased by the fact that the user device has moved out of range before it had the time to receive the beacon frame and send an answer.

According to all of this, it can be concluded that the greatest benefit when counting passengers in PPT vehicles is obtained by implementing an APC solution based on the probe request processing and analysis. The APC solution that uses probe request frames (active scanning) is currently the best usage of Wi-Fi technology since sending a frame is faster than sending a beacon and has enough information in the said frame to determine the number of user devices. The architecture of the system based on Wi-Fi consists of several components, but the counting concept is the same for most frames. The logical representation of the Wi-Fi-based APC can be seen in Fig. 2.

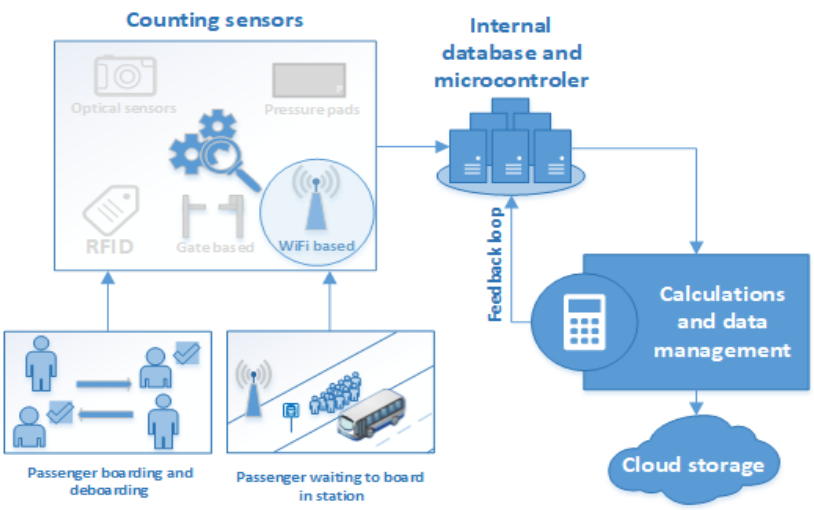

Fig. 3. Architecture of Wi-Fi-based APC solution

With everything already described, the user devices transmit probe request frames for the purpose of finding wireless networks that they can connect to. Generally, the components detect the frame and store it in an internal or external database. Once the frames are stored they must be filtered and calculated in a data management system. After processing, the results are stored (generally in some cloud storage system) and sometimes used in a feedback loop to calibrate the components for the future measurements. The stored data should be ready and accessible when requested for analysis and decision-making process.

It is important to note that no APC solution or counting method provides 100% reliability or precision and the main disadvantage, as mentioned earlier, is the difference in the number of entries and exits to and from the vehicle. However, APC solutions based on Wi-Fi technology are distinctive because of their, already described, core concept which is based on in-vehicle device detection. Taking this into consideration, there are numerous parameters that can affect the reliability and precision of the counting results that are specific to their Wi-Fi base and are the reason for the aforementioned calibration. Fig. 4 shows the concept of probe request frame collection and is based on logical representation for most APC solutions, as shown in Fig. 3.

For the stated reasons, a precise calibration is necessary and must take into account the deviations from the actual number of passengers and the signal strength in the vehicle so that it can compensate for the difference. This procedure connotes a reliable process for verification of the collected data and calculated or estimated results.

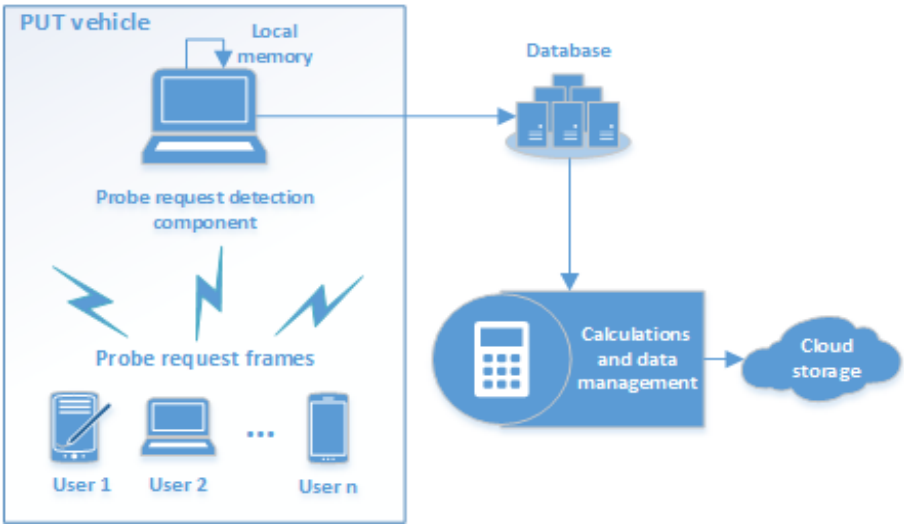

Fig. 4. Concept of probe request-based counting

An example of a calibration process is the use of manual counting methods or some other form of automatic counting that can be used to compare the results. This, however, creates a hybrid variation of the proposed solution, but is most effective when it

comes to the precision of the results. Some other methods for "*cleaning*" the collected data are certain statistical models implemented in the used software to analyze and process the data (for example, counting how many times a unique MAC address is repeated, or defining a signal strength threshold to determine if the device/person is inside the designated counting zone).

4 Analysis of the existing research results

Consequently to everything already stated in this research, Wi-Fi passenger counting uses the gathering and processing of data obtained from the probe request framework as the basis for counting user devices/passengers and the methods can differ because of the different conditions and possibilities in which the authors conducted their experiments. Therefore, this section presents an overview of the related literature and the research done on Wi-Fi-based APC and the analysis of the results of specific existing research from the field of APC solutions based on Wi-Fi. An overview of the studies related to the solution of Wi-Fi-based APC in public urban transport is shown in Table 1. Columns in the table represent what was considered important information of the researched papers.

Table 1. Overview of studies related to the solution of Wi-Fi based Automatic Passenger Counting in Public Urban Transport.

Authors	**Mode of transport**	**Measuring device location**		**Estimation an calculation**	**Counting performed**	**Methods**
		Inside	**Station**			
Seunghan et al. (2020) [6]	Bus	-	+	OD matrices, waiting times	+	Probe request frame collecting, Wi-Fi signal strength and video recording analysis
Hidayat et al. (2020) [8]	Bus	+	-	OD matrices, mean absolute error, median absolute error, standard deviation of absolute error, root mean square, error, R2 score	+	Probe request frame collecting

Ribeiro et al. (2020) [9]	Bus	+	-	Number of passengers, peak values and OD matrices	+	Probe request frame collecting
Nitti et al. (2020) [23]	Bus	+	-	Number of passengers, bus running time	+	Probe request frame collecting
Moser et al. (2019) [24]	Bus	+	-	Number of passengers	+	Probe request frame collecting
Oliveira et al. (2019) [16]	/	-	-	Mean relative error	+	Probe request frame collecting
Hakegard et al. (2018) [7]	Bus	+	-	OD matrices	+	Probe request frame collecting
Das et al. (2018) [11]	/	-	-	Passively encountering group (PEG)	+	Probe request frame collecting
Hidayat et al. (2018) [25]	Bus	+	-	Non-passenger data	+	Probe request frame collecting
Ribeiro et al. (2017) [26]	Bus	+	+	OD matrices, number of passengers	+	Probe request frame collecting
Song et al. (2017) [27]	Train	-	-	Waiting times, spectral clustering approach	-	Probe request frame collecting
Song et al. (2017) [13]	/	-	-	Length of stay (LOS), passenger flow and occupancy	+	Probe request frame collecting
Hakegard et al. (2017) [28]	Bus	+	-	Number of passengers	+	Probe request frame collecting, signal strength analysis
Kang et al. (2016) [29]	Bus	+	-	OD matrices, mobile device tracking	+	Probe request frame collecting
Oransirikul et al. (2014) [15]	Bus	+	+	Number of passengers	+	MAC address collecting

The first column contains the name of the author, year of publication and the referenced paper. The second column represents the mode of transport on which the particular solution was based (bus, train or none if in laboratory conditions) and the third column, split in two, was created to show where the equipment was implemented (inside or outside the vehicle). Next two columns represent the results the authors estimated and/or calculated in their papers and the last column provides an overview of the used methods. Therefore, Table 1 demonstrates how many different variants are possible when using Wi-Fi as the basis of the APC solution.

As described in the previous sections, the core concepts of collecting probe request frames or measuring the signal strength can deliver datasets that can be used for any number of different possibilities - ridership, timestamps, OD matrices, waiting times, boarding and deboarding, peak times and values, length of stay, passenger grouping, mean absolute error, median absolute error, etc. So, alongside ridership data can be used for generating a lot of other information and any form of PmA to optimize, modernize or enhance the already existing PPT systems or help in the creation of a new one.

Most research that was analyzed was conducted in buses, some were conducted in trains/metro and some in laboratory conditions at universities or some other enclosed environment (office, large hall, etc.). It is not necessary to state that the experiments in dynamic real life conditions, in PPT systems and vehicles [6], [7], [8], [9], [15], [23], [24], [25], [26], [27], [28], [29] are more relevant and have higher reliability status than those that were conducted in the controlled conditions with volunteers [10], [11], [12], [13], [14], [16], [17].

However, the aim of this research is to demonstrate that there are many different methods and analytical processes for collecting and interpreting the gathered data from the Wi-Fi network. This information might not be useful for some developed countries that already have some form of APC solution integrated in their PPT system, but it can serve as a great overview for less fortunate transport service providers to help them choose a method for counting passengers using their existing Wi-Fi network. Also, this can be of great help to other researchers, who are about to start experimenting in this field, to easily find some relevant literature on the subject and review the already explored methods and processes.

5 Conclusion

Counting passengers using various technologies and methods, including Wi-Fi, is being increasingly introduced in the running and maintaining the public transport systems, networks and vehicles. By analyzing the available literature and research, it was found that there is currently no clear taxonomy of selecting the optimal method or methodology for counting passengers using Wi-Fi technology. The main difference is in the approach and is shown depending on the location of the counting equipment, the type of vehicle and mode of transportation, the estimated results and actual measured values, the chosen method for filtering and analyzing the gathered raw data, etc.

An important role in the APC solutions based on Wi-Fi technology has the understanding of the scanning methods and the difference between active and passive scanning and counting by different authors, which can be characterized as a potential challenge and a problem when it comes to terminology and taxonomy. This is why the authors of this paper made a decision to classify all the unobtrusive, non-invasive and indirect methods as passive from the perspective of counting passengers in PPT.

In the future research, the authors expect to conduct their own pilot experiment for counting passengers, i.e. counting in a real dynamic traffic environment (PPT vehicle in motion, at a stop, at the terminal turn stop, in a garage, etc.) oriented to active and passive scanning using Wi-Fi technology. The planned experiment will be conducted in the city of Zagreb in Zagrebački Električni Tramvaj – ZET (Zagreb Electric Tramway), transport network and public transport vehicles.

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